PAYTM BUILD FOR INDIA AI HACKATHON · BENGALURU EDITION | TRACK 1

PaytmAI VyaparVoice

The Proactive Voice Growth, Operations & Retention Autopilot for Every Paytm Merchant

Transforming 1.57 Crore Paytm Soundboxes from passive payment speakers into active vernacular business partners — turning UPI history into verified repeat sales, smarter stock prep, and happier customers in 60 seconds a day.

01 · Vernacular Voice-First

Kannada, Hindi & English closing pitch; zero typing

02 · Guardrailed by Design

DPDP consent, ₹300 hard cap & ≤22% margin floor

03 · Causal Holdout ROI

Built-in 80/20 control split proving 5.8x net ₹ lift

TEAM: SemiTechnical

Builder: Kevin Joseph · Bengaluru, India

Track 1: Merchant Growth AI
Working MVP Prototype Included

PAYTM AI SOUNDBOX · 8:30 PM

LCD · SRI KRISHNA DARSHINI

+₹1,740 Net Sales Lift

40 Regulars Eligible · ₹300 Max Cap

▸ EVENING KANNADA / EN BRIEF

"Namaskara Ramesh avare! 40 regular evening buyers haven't visited in 18 days. Send ₹25 UPI cashback on ₹150+ bill? Max spend locked at ₹300."

▸ *Merchant Reply: "Haudu, kaluhisi!"*

✔ **WORKING PROTOTYPE BUILT**

FastAPI + SQLite + Indic Voice +
5-Rule Guardrails + Holdout Engine

1.57 Cr Merchants Accept Digital Payments — Yet Run Blind on Retention & Ops

Paytm digitized payment collection across India, but small shopkeepers still lose 25–35% of repeat buyers silently.

1. Silent Repeat-Customer Churn (28–35% Annual Attrition)

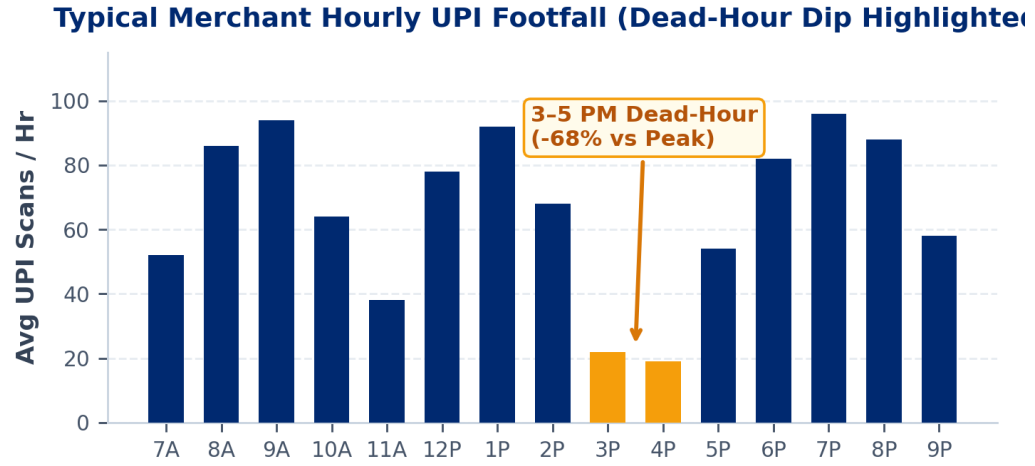
A busy shop processes 150–300 UPI scans daily, but treats every QR payment as an isolated alert. Merchants never notice when loyal ₹2,000/month regulars quietly stop visiting until monthly cashflow drops.

2. Zero Time or Literacy for BI Dashboards & Chat Prompts

Kirana, darshini, and pharmacy owners stand at busy counters 12+ hours/day. Existing 'analytics apps' fail (<8% WAU) because merchants won't type prompts, export CSVs, or configure CRM rules.

3. Predictable Dead Hours & Unplanned Rush Stockouts

Stores suffer a 60–70% footfall drop every afternoon (3–5 PM) causing perishable waste, yet run out of high-velocity staples (milk, batter) on payday weekends due to guesswork ordering.



GROUND REALITY · BENGALURU MERCHANT PERSONA

Ramesh Rao · Sri Krishna Darshini & Provisions (Indiranagar)
Processes ~917 Paytm UPI scans/month (₹1.75L GMV), yet had 48 regular evening buyers go silent for 18+ days without realizing it.

1.57 Cr	₹7.1L Cr	< 7%
Paytm Device Shops	Quarterly Merchant GMV	Current Loan Penetration

PaytmAI VyaparVoice: Proactive Voice Partner on Every Counter

Directly engineered around all 3 pillars of Track 1 — Grow Business, Manage Operations, and Serve Customers Better.

PILLAR 1 · GROW BUSINESS

Win-Back & Dead-Hour Autopilot

- **Silent Churn Radar:** Flags high-frequency UPI payers inactive for 15–30 days automatically.
- **Dead-Hour Flash Boost:** Fills 3–5 PM slow slots with hyperlocal combo nudges to afternoon buyers.
- **1-Tap UPI Cashback Pass:** Auto-redeemed on next Paytm QR scan — zero paper coupons or cashier friction.
- **Causal Holdout Proof:** Holds 20% silent control group to prove exact net incremental ₹ earned.

PILLAR 2 · MANAGE OPERATIONS

Predictive Rush & Supplier Copilot

- **Payday & Weekend Forecasting:** Detects +28% end-of-month & weather spikes from historical scan velocity.
- **1-Command Supplier PO:** Sends pre-filled WhatsApp restock orders (milk, batter, staples) to local vendors.
- **Perishable Margin Guard:** Triggers evening clearance bundles on slow-moving fresh stock before closing.
- **Working-Capital Readiness:** Tracks cashflow & EDI coverage to unlock pre-approved Paytm partner credit.

PILLAR 3 · SERVE CUSTOMERS

Vernacular Post-Visit Delight

- **Queue-Delay Detection:** Spots unusually slow peak-hour clusters & 3-star Paytm payment feedback tags.
- **Instant Goodwill Recovery:** Dispatches merchant-signed Kannada/Hindi apology + ₹30 coffee pass in 1 tap.
- **Zero-App Loyalty Milestones:** Automatically celebrates a regular's 10th UPI scan on Soundbox audio.
- **DPDP-Compliant Privacy:** Strictly honors customer opt-in & enforces 14-day anti-spam cooldowns.

CORE DESIGN PRINCIPLE: "ONE EVENING VOICE ACTION — NEVER ANOTHER DASHBOARD"

Instead of expecting shopkeepers to analyze graphs, VyaparVoice speaks ONE high-confidence recommendation at 8:30 PM closing time in Kannada, Hindi, or English. The merchant simply says "Haudu / Haan, bhej do" — and AI executes end-to-end.

60 Seconds at Closing Time: From UPI Signal to Verified ₹1,740 Net Lift

End-to-end workflow already running in our Working MVP Prototype (Sri Krishna Darshini, Indiranagar dataset).

STEP 1 · 8:25 PM

Zero-Entry UPI Scan

SQLite/Postgres RFM engine scans 1,613 Paytm QR txns across 240 payers. Flags 48 evening regulars silent for 16–28 days.

→

STEP 2 · 8:29 PM

5-Rule Policy Guardrail

Filters to 40 DPDP-consented & cooldown-clear users. Verifies ₹25 off ₹150+ (16.7% ≤ 22% margin cap) & locks ₹300 max spend.

→

STEP 3 · 8:30 PM

Vernacular Voice Pitch

Soundbox speaks pitch in Kannada/Hindi/English. Merchant replies "Haudu, approve!" (or 1-tap). 32 nudged; 8 held in silent Control.

→

STEP 4 · +48 HOURS

Causal ROI Verification

12/32 nudged regulars scan & pay (37.5%) vs 1/8 Control (12.5%). Soundbox reports +₹1,740 net new sales on ₹300 spend (5.8x ROI).

LIVE WORKING PROTOTYPE INCLUDED · REAL FASTAPI + SQLITE COMPUTATION (ZERO HARDCODED MOCK NUMBERS)

01 · Interactive AI Soundbox Simulator

- Live switch between English, Kannada (KN), and Hindi (HI) evening voice briefings.
- Browser SpeechSynthesis + Microphone voice intent recognition ('Haudu / Haan, approve!').
- Animated Soundbox LED + real-time LCD status readout.

02 · Deterministic Policy Inspector

- Displays exact pass/fail counts on every campaign card before merchant approval.
- Enforces DPDP opt-in (43/48), 14d cooldown (40/43), ≤22% discount floor, and ₹300 hard liability cutoff.

03 · 48-Hr Time-Travel & Ops Engine

- 1-click simulation inserts redemption UPI scans into SQLite & computes causal lift vs 20% holdout.
- Includes Weekend Dairy Supplier WhatsApp PO (+18L milk) & Queue-Delay Goodwill Recovery.

Low-Latency, Guardrailed Agentic Architecture Built for Vernacular Scale

LLMs handle natural vernacular conversation; deterministic Python engines govern money, consent, and math.

LAYER 1 · MERCHANT & CUSTOMER TOUCHPOINTS	**Paytm AI Soundbox 4G** Scheduled evening audio briefing + single-button / voice confirmation	**Paytm for Business PWA** Lightweight React/HTML5 visual companion card & live ROI tracker	**Paytm App & WhatsApp Cloud** Instant UPI Cashback Pass delivery & 1-tap supplier WhatsApp POs
LAYER 2 · INDIC VOICE & AGENTIC INTELLIGENCE LAYER	**Sarvam AI + Web Speech** Natural Kannada, Hindi, Tamil & Indian-English ASR + TTS (<400ms latency)	**SLM / LLM Intent Router** Llama-3.3 / Qwen structured tool-calling via Groq for ultra-low inference cost	**Semantic Store Memory** Cognee / Vector store retaining merchant margin preferences & local calendar
LAYER 3 · DETERMINISTIC POLICY, RFM & CAUSAL ENGINE (CORE)	**Python FastAPI Core** Async microservices computing RFM cohorts & hourly velocity troughs	**5-Rule Policy Engine** Hard deterministic checks: DPDP consent, 14d cooldown, margin floor & ₹ cap	**80/20 Holdout Attribution** Randomized control split & redemption matcher calculating true incremental ₹
LAYER 4 · TRANSACTION DATA & EVENT ORCHESTRATION	**PostgreSQL / SQLite Ledger** Paytm QR UPI scan stream, customer consent state & immutable audit trail	**Redis Coupon Counter** Sub-millisecond atomic redemption counter enforcing instant ₹300 budget cutoff	**n8n Workflow Automation** 8:30 PM closing cron triggers, 48h ROI report callbacks & supplier alerts

Why VyaparVoice Wins Where Dashboards & Generic AI Chatbots Fail

Purpose-built around Paytm's unfair hardware + UPI network moat and Indian shopkeeper behavior.

Dimension	Traditional BI / Khata Apps	Generic 'Chat-With-Data' AI	PaytmAI VyaparVoice (Ours)
Merchant Effort	Manual entry + complex charts (<8% weekly active usage)	Requires typing prompts & asking the right business question	**Proactive 20-sec Kannada/Hindi Soundbox briefing + 'Haan' voice approval**
Customer Data Source	Manual phonebook / incomplete paper bills	Disconnected from real payment identity	**Zero-entry Paytm QR repeat-payer graph across 1.57 Cr storefronts**
Financial & Legal Safety	Static rules; manual coupon abuse risk	LLM hallucination risk on discounts & spam violations	**Deterministic 5-rule policy engine (DPDP opt-in, margin cap, ₹300 cutoff)**
Action Execution	Passive insights only — merchant must act alone	Text suggestions only; no payment rail hook	**Closed-loop Paytm UPI Cashback Pass + 1-tap WhatsApp Supplier PO**
Proof of ROI	Vanity SMS sent / message open counts	No causal measurement	**Built-in 80/20 Holdout Control Split proving true net incremental ₹**

1. Hardware Distribution Moat

Rides on Paytm's deployed 1.57 Cr Soundboxes — zero extra hardware capex or new app install.

2. Zero-Friction Redemption

Cashback unlocks automatically when the customer scans the merchant's Paytm QR — zero cashier training.

3. Trust Through Honest Math

Showing holdout-verified net ₹ lift builds lasting merchant trust to renew SaaS every month.

Win-Win Flywheel: Verified Rupee Profit for Shops & Stickiness for Paytm

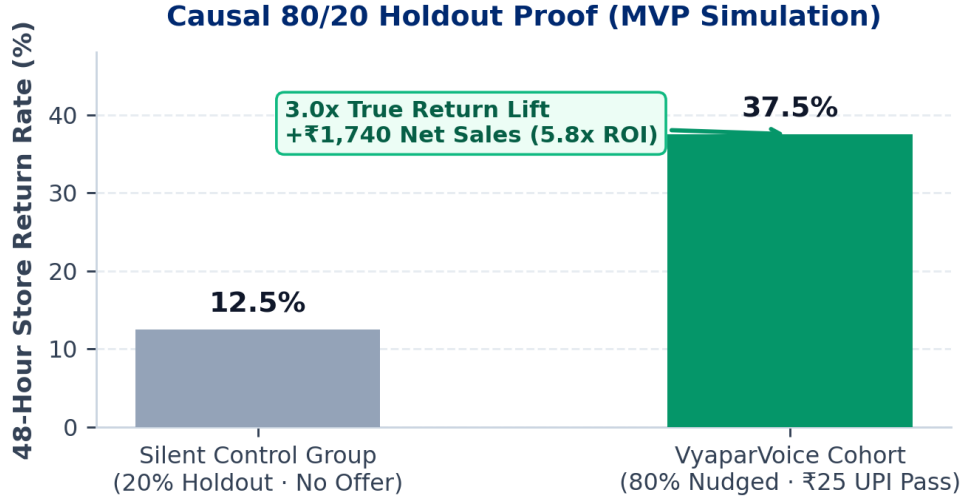
Every incremental repeat visit grows merchant income while compounding Paytm's GMV, device ARPU, and credit data.

FOR PAYTM MERCHANTS (1.57 CR SMBs)

- **+14% to +22% Repeat Visits:** Recovers silent churned regulars before they switch permanently to competitors.
- **5.8x Verified Campaign ROI:** +₹1,740 net incremental sales on ₹300 capped spend in 48h MVP simulation.
- **< 60 Seconds Daily Effort:** 100% hands-free Kannada/Hindi voice workflow while closing the shutter.
- **-18% Dead-Hour & Stock Loss:** Afternoon combo nudges + predictive weekend dairy/staple POs protect margins.

FOR PAYTM ECOSYSTEM (STRATEGIC FLYWHEEL)

- **-35% Soundbox Churn:** Transforms Soundbox from a ₹100/mo cost center into a visible profit-generating partner.
- **+12–15% Store QR GMV:** Cashback passes require payment via Paytm UPI QR, capturing greater counter share.
- **7% → 15% Lending Funnel:** Verified retention stability & cashflow curves unlock low-risk merchant loan distribution.
- **Consumer App Stickiness:** Hyperlocal rewards from neighborhood darshinis & kiranas boost daily Paytm app opens.



5.8x
Avg Net Campaign ROI

+79%
Soundbox ARPU Lift

< ₹2
Monthly AI Cost / Shop

Zero-CAC SaaS + Pay-for-Success Take-Rate Scaling to ₹243 Cr+ ARR

Monetizes Paytm's existing 1.57 Cr merchant billing rail with >78% software gross margins.

1. Soundbox 'Growth Pro' SaaS Add-On — ₹79 / Month

Bundled onto existing ₹100/mo Soundbox subscription (+79% device ARPU uplift) with near-zero customer acquisition cost. First 2 win-back campaigns free to prove ROI before billing.

2. Pay-for-Success Redemption Micro-Fee — 5% of Attributed Promo Bill

Charged ONLY when a nudged customer scans & pays at the counter (e.g., ~₹9.50 fee on a ₹190 recovered bill). 100% risk-free alignment for small shopkeepers.

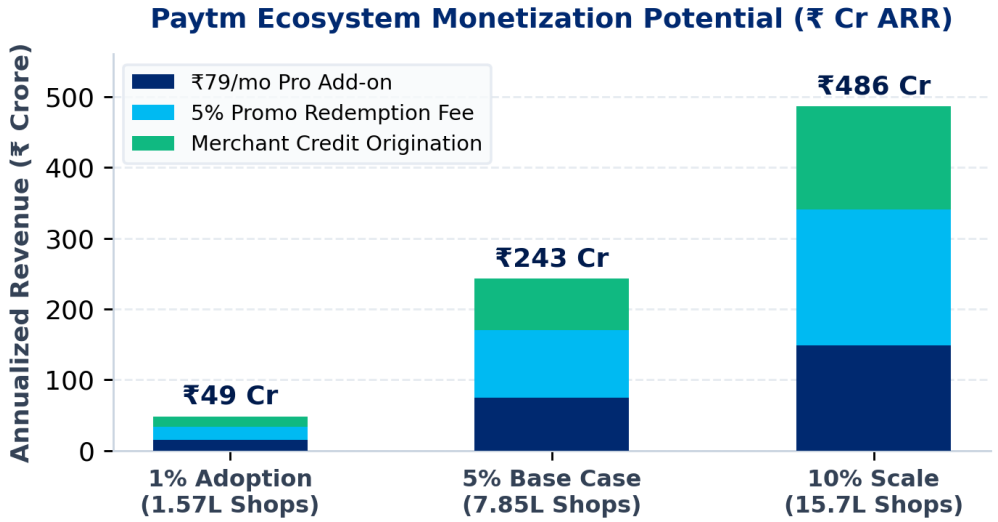
3. Merchant Lending Origination Uplift — 1.5% to 2.5% Partner Fee

High-retention merchants growing GMV via VyaparVoice pre-qualify for Paytm's asset-light NBFC/bank working-capital loans, expanding Paytm's highest-margin revenue engine.

UNIT ECONOMICS PER ACTIVE MERCHANT / MONTH (~81% GROSS MARGIN)

Blended Monthly ARPU: ₹258/shop (₹79 SaaS + ₹102 Promo Fee + ₹77 Credit Uplift)

Monthly COGS: ~₹48/shop (SLM inference ₹1.80 + Cached Indic TTS ₹3.20 + WhatsApp/Push ₹43)



BASE CASE AT JUST 5% DEVICE ADOPTION (7.85 LAKH SHOPS)

₹243 Crore High-Margin ARR

Directly supports Paytm's AI commerce & cloud monetization roadmap with zero balance-sheet credit risk.

07 · EXECUTION ROADMAP & LIVE DEMO

Built Today, Engineered to Scale to Millions of Storefronts

From working Bengaluru hackathon vertical slice to pan-India 11-language Soundbox deployment.

PHASE 1 · BUILT TODAY (MVP)

Working Vertical Slice Ready

✔ Real SQLite UPI ledger (240 payers, 1,613 scans) + FastAPI RFM & hourly trough engine

✔ Kannada, Hindi & English AI Soundbox voice simulator + speech command approval

✔ 5-rule deterministic guardrail engine + 48h causal holdout ROI simulator (5.8x ROI)

PHASE 2 · DAYS 1–45 PILOT

Bengaluru 1,000-Store Live Beta

✔ Deploy across 1,000 F&B, kirana & pharmacy Soundbox merchants in Bengaluru

✔ Integrate live Sarvam AI streaming audio + Paytm UPI Dynamic Offer Ledger

✔ Validate 30-day repeat GMV lift, coupon auto-cutoff & merchant renewal conversion

PHASE 3 · DAYS 45–120 SCALE

Pan-India 11-Language Rollout

✔ Expand across all 11 Soundbox languages on Paytm's 1.57 Cr merchant device fleet

✔ Launch hyperlocal distributor B2B restock marketplace & FMCG brand-funded offers

✔ Feed verified retention scores into Paytm Intelligence (Pi) credit distribution

PaytmAI VyaparVoice · Every Merchant's AI Growth Partner

Paytm Build for India AI Hackathon – Bengaluru Edition | Track 1: Merchant Growth AI

Team: SemiTechnical
Builder: Kevin Joseph
Status: Pitch + Live MVP Ready